Interference-Based Quantum Classification of Histopathology Images Using Linear State Overlap

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***Abstract*—Near-term quantum machine learning (QML) is constrained by two practical bottlenecks: encoding high-dimensional medical images into quantum states and performing measurement-intensive inference when decisions rely on quadratic fidelities. We address this gap with a hybrid medical-QML pipeline in which a classical CNN backbone with a compact neck compresses 96×96 lymph node histopathology patches into a low-dimensional embedding that is amplitude-encoded, followed by a fixed interference circuit that classifies using the *sign of the real linear overlap*. Learning is performed entirely in the classical domain: class-representative prototype states are constructed by centroid aggregation of embeddings and compiled into fixed state-preparation unitaries, enabling inference with one fixed circuit per sample and a small constant number of shots. Experimental evaluation uses the PatchCamelyon dataset with statevector simulation and realistic noise injection. On the PCam validation subset, the proposed IQC achieves 88.07% accuracy while maintaining measurement-efficient, low-shot inference. These results indicate a practical NISQ-compatible classifier tailored for medical image inference under resource constraints.**

***Index Terms*—NISQ Devices, Quantum Machine Learning, Quantum Classification, Quantum Interference, Linear Overlap, Hybrid Quantum–Classical Systems, Medical Imaging, Histopathology**

I. Introduction

A. Background of Quantum Machine Learning

Quantum machine learning (QML) promises new modeling primitives, but near-term classifiers remain constrained by either unstable variational training or measurement-intensive similarity estimation [1]–[3], [6]. These challenges intensify in medical imaging, where raw lymph node histopathology patches are high-dimensional and direct quantum encoding of 96×96 images is infeasible on NISQ hardware.

B. Limitations of Existing QML Approaches

Current QML pipelines often rely on parameterized circuits that are difficult to train at scale or on similarity estimation procedures that demand large numbers of measurement shots. In clinical imaging tasks, both training stability and measurement efficiency are critical to feasible deployment.

C. Motivation for Interference-Based Classification

We address this gap with a hybrid medical-QML architecture. A classical CNN backbone with a compact neck compresses each image into a low-dimensional embedding. The embedding is then amplitude-encoded into a quantum state and processed by a fixed interference circuit. Classification is performed by the sign of the real linear overlap between the test state and a class prototype, requiring a small constant number of shots per inference using one fixed circuit per sample. Learning is entirely classical: class-representative prototype states are constructed by centroid aggregation of embeddings, and the quantum processor is used only for inference.

D. Contributions of This Work

Our contributions are:

- A **hybrid medical-QML pipeline** for lymph node histopathology that makes quantum encoding feasible by using a CNN neck for dimensionality reduction.
- A **linear-interference decision primitive** based on $\mathrm{sign}(\mathrm{Re}\langle\phi|\psi\rangle)$, yielding a hyperplane classifier in Hilbert space.
- A **fixed interference circuit** requiring measurement-efficient inference with a small constant number of shots and no variational training.
- A **classical learning phase** using centroid prototypes, decoupling training from quantum execution, with experimental validation achieving 88.07% accuracy on the PCam validation subset.

E. Structure of the Paper

The remainder of this paper is organized as follows. Section II reviews the relevant QML literature. Section III summarizes existing system paradigms and their limitations. Section IV presents the proposed interference-based classifier, including the hybrid pipeline, encoding, and decision mechanism. Section V details the implementation setup, datasets, and evaluation protocol. Section VI discusses measurement efficiency, noise robustness, limitations, and scalability. Section VII concludes and outlines future work.

II. Literature Review

Quantum machine learning (QML) has progressed from early linear-algebraic subroutines to hardware-focused models tailored to NISQ constraints. We summarize the main approaches and position the linear-interference framework within the remaining gaps.

A. Variational Quantum Classifiers

Variational quantum classifiers are the most well-known type of machine learning model from the NISQ era [1], [5]. VQCs use a parameterized quantum circuit (PQC) as an ansatz, and they change the parameters to make a classical cost function as small as possible. VQCs are very adaptable, but they are also very hard to train. Gradient-based optimization is often not possible for larger systems because there are empty plateaus in the parameter space where gradients disappear exponentially with the number of qubits [2]. Also, VQCs need a lot of measurement shots at each optimization step to get low-variance gradient estimates, which puts a lot of strain on the quantum processor's duty cycle.

B. Quantum Kernel and Similarity-Based Methods

Quantum kernel methods circumvent the optimization challenges associated with VQCs by projecting data into high-dimensional Hilbert space and calculating the similarities between pairs of data points [3], [6]. These similarities, commonly characterized as the fidelity $F(\rho, \sigma) = \mathrm{Tr}(\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}})^2$, serve as kernel entries in classical algorithms such as Support Vector Machines (SVMs). Kernel approaches are attractive but can be measurement-intensive on quantum hardware. To classify a new point against a dataset of size N, one typically requires $O(N)$ similarity measurements. Fidelity is also a quadratic observable that discards relative phase information, which can matter when the decision boundary depends on sign structure in the amplitudes.

C. Quantum State Discrimination and Metric Learning

Quantum state discrimination (QSD) methodologies regard classification as the process of determining which of several known quantum states a particular system occupies [7], [8]. From this point of view, the focus moves from circuit parameters to the way states are arranged in Hilbert space. Quantum metric learning, on the other hand, tries to find a feature map that makes the distance between states that represent different classes as large as possible and the distance between states that represent the same class as small as possible. Our research is based on these geometric principles, but we add a new way of making inferences: we use a fixed interference circuit to get a linear overlap signal, which is a simpler and more efficient way to measure than full state tomography or fidelity estimation.

D. Interference-Based Quantum Classification

Works like the distance-based classifier by Schuld et al. [4], [11] pioneered the use of the Hadamard test for pattern recognition by calculating distances between vectors, and kernel-based variants compute signed fidelity sums in superposition [9], [10]. Our work differs operationally in two ways: (i) the decision variable is the **sign of the real overlap**, not a fidelity or distance estimate, and (ii) learning uses a classical centroid construction, with a **fixed interference circuit** used only at inference. This yields a measurement-efficient primitive that avoids variational training and reduces circuit executions per sample.

III. Existing System

A. Variational Quantum Classifiers

Variational quantum classifiers (VQCs) use a parameterized quantum circuit whose parameters are tuned to minimize a classical loss. They are expressive, but training requires iterative quantum-classical loops with many shots to estimate gradients, making them resource-intensive on NISQ devices.

B. Quantum Kernel-Based Classifiers

Quantum kernel methods embed data into a high-dimensional Hilbert space and compute pairwise overlaps that serve as kernel values for classical SVMs. They avoid variational training but require repeated overlap estimation during inference, which can be measurement-intensive as dataset size grows.

C. Fidelity-Based Quantum Similarity Methods

Fidelity-based schemes measure similarity using $F(\phi, \psi) = |\langle\phi|\psi\rangle|^2$. This is robust to noise but discards phase information and yields quadratic decision boundaries, often requiring many shots to estimate accurately.

D. Challenges in Existing Systems

Existing models exhibit three recurring limitations: (i) variational training suffers from barren plateaus and high-variance gradient estimates, (ii) kernel and similarity-based approaches incur measurement-intensive overlap estimation, and (iii) fidelity-based similarity discards the sign of interference, which can be informative for decision boundaries. These issues motivate alternatives that preserve interference structure while reducing measurement overhead.

E. Limitations of Existing Systems

Across these paradigms, the main inefficiencies arise from optimization instability (VQCs), large measurement budgets (kernel and fidelity methods), and loss of sign information in quadratic overlaps. These limitations motivate a linear-interference decision rule that retains phase-sensitive information while using a fixed, low-shot circuit.

IV. Proposed Work

A. Overview of Hybrid Pipeline

The proposed Interference-Based Quantum Classifier (IQC) follows a modular pipeline: *Input image* $\rightarrow$ *CNN backbone* $\rightarrow$ *compact neck embedding* $\rightarrow$ *quantum state-preparation* $\rightarrow$ *interference-based decision*. The classical module compresses high-dimensional images into an embedding compatible with a small qubit register, while the quantum module performs a

TABLE I
COMPARISON OF QUANTUM CLASSIFICATION PARADIGMS

Feature	Variational (VQC)	Kernel (QSVM)	Fidelity-Based	**Linear Interference (IQC)**
Decision Basis	Probabilities $P(y\|x)$	High-dim Kernels	Quadratic Overlap F	**Linear Overlap** $\mathrm{Re}\langle\phi\|\psi\rangle$
Learning Mode	In-circuit Optimization	Classical SVM ($O(N^2)$)	Static/Adaptive	**Static Classical Aggregation**
Quantum Circuit	Parameterized (Trainable)	Fixed (Feature Map)	Fixed (Swap Test)	**Fixed (Hadamard Test)**
Shot Complexity	High ($O(M \cdot N)$)	Moderate ($O(N)$)	Moderate ($O(1/\epsilon^2)$)	**Low (constant shots)**
Phase Sensitivity	No	No	No	**Retains sign of real interference**
Optimization Risk	Barren Plateaus	None (in quantum)	None	**None**

fixed interference measurement to obtain a sign-based decision.

Training is executed classically. We first train the CNN backbone and neck on PCam using standard cross-entropy loss to produce discriminative embeddings. After training, the CNN+neck is frozen and used to generate embeddings for all training samples. These embeddings are aggregated into class prototypes (Section IV-F), which are compiled into fixed state-preparation unitaries. Quantum hardware is used only for inference: for each test sample we compute its embedding, amplitude-encode it as $|\psi\rangle$, run the fixed interference circuit against the class prototype, and assign a label based on the sign of the interference score estimated over a small constant number of shots.

Fig. 1 illustrates the complete hybrid classical–quantum pipeline used in the proposed IQC framework.

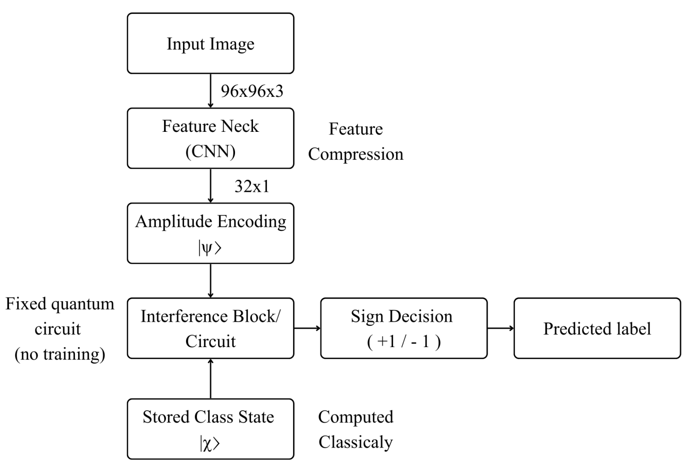

Fig. 1. Flowchart of the hybrid classical-quantum inference pipeline using the Interference-Based Quantum Classifier (IQC).

B. Classical Feature Extraction Module

The hybrid pipeline described in Section IV-A begins with a classical CNN backbone that processes 96×96 RGB lymph node histopathology patches and extracts hierarchical features correlated with malignancy. To make quantum encoding feasible, we insert a compact "neck" module that compresses the feature maps into a low-dimensional embedding. The neck acts as a learned projection (analogous to a task-specific PCA), implemented with global average pooling followed by a fully connected layer. The embedding dimension d is chosen to match the available qubit budget via $n = \lceil \log_2 d \rceil$. Fig. 2 provides a visual representation of the backbone and feature neck.

1) Module-wise Tensor Shapes: The end-to-end data flow is:

- Input image: $(96, 96, 3)$
- CNN feature maps: (h, w, c) (architecture-dependent)
- Global average pooling: (c)
- Neck projection: (d)
- Quantum feature vector (normalized and zero-padded if needed): (2^n)
- Quantum register: n data qubits + 1 ancilla
- Output: 1 interference score (sign used for label)

This structure explicitly defines the image $\rightarrow$ neck $\rightarrow$ quantum classifier tail, ensuring the quantum stage operates on a manageable number of qubits while preserving discriminative features.

C. Quantum Encoding Module

A critical component of the IQC is the explicit mapping of classical feature embeddings into quantum states. We adopt **amplitude encoding** as the primary feature map $\Phi : \mathbb{R}^d \rightarrow \mathbb{C}^{2^n}$. A classical embedding $\mathbf{x} \in \mathbb{R}^d$ is first normalized and (if needed) zero-padded to length 2^n, then mapped to an n-qubit state via a state-preparation unitary $U(\mathbf{x})$:

$$|\psi\rangle = \sum_{i=0}^{2^n-1} x_i |i\rangle, \tag{1}$$

with $n = \lceil \log_2 d \rceil$. Amplitude encoding provides the highest information density, encoding d features into n qubits. In our implementation, amplitude encoding is realized via Qiskit's `StatePreparation` gate, which constructs a sequence of uniformly controlled rotations to prepare any normalized complex vector $\mathbf{x}$ as a quantum state $|\psi\rangle$. The interference mechanism itself is independent of the specific compilation used.

We consider three primary encoding strategies adapted for the IQC framework:

1) **Amplitude Encoding**: Maps a normalized vector $\mathbf{x} \in \mathbb{R}^d$ to the amplitudes of a 2^n-dimensional quantum state. This provides the highest information density ($O(2^n)$ features) but requires complex state-preparation circuits. In IQC, amplitude encoding ensures that the linear

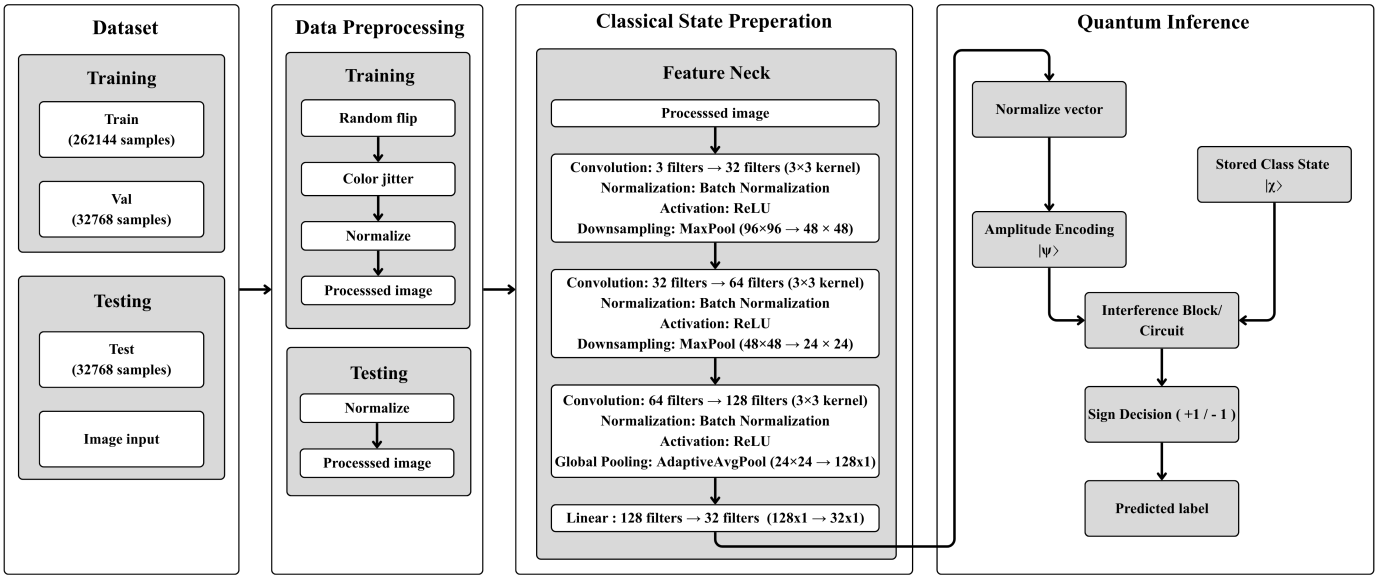

Fig. 2. Hybrid pipeline: image → CNN backbone → neck (dimensionality reduction) → quantum encoding → interference-based classifier tail. The quantum stage uses n data qubits and one ancilla for sign-based inference (small constant shots).

overlap in Hilbert space is directly proportional to the Euclidean dot product in the input space.

2) **Angle Encoding**: Maps each feature x_j to the rotation angle of a qubit, $|\psi\rangle = \bigotimes_j (\cos x_j |0\rangle + \sin x_j |1\rangle)$. This is hardware-efficient but introduces nonlinearities (trigonometric functions) into the decision signal.
3) **Basis Encoding**: Maps classical bitstrings directly to computational basis states. While simple, it does not exploit the superposition-based interference that defines the IQC's advantage.

In this work, we use amplitude encoding for all reported experiments.

1) Linear Separability in High-Dimensional Feature Spaces: According to Cover's Theorem [17], a complex pattern classification problem cast in a high-dimensional space nonlinearly is more likely to be linearly separable than in a low-dimensional space. By mapping data into the 2^n-dimensional Hilbert space, the IQC leverages this high dimensionality to separate classes using a simple hyperplane.

However, unlike kernel methods that use an implicit feature map, the IQC works with an **explicit feature map** $\Phi(\mathbf{x})$. The linear overlap $\mathrm{Re}\langle\phi|\psi\rangle$ is computed directly in the state space. The success of the static centroid construction (Section IV-F) depends on whether the chosen Φ places the class distributions in convex regions that can be separated by a single hyperplane.

D. Linear Interference Decision Mechanism

The central theoretical contribution of this work is the application of linear interference as the main signal for quantum classification.

1) Linear vs. Quadratic Similarity: In conventional schemes, similarity is often expressed through fidelity $F(\phi, \psi) = |\langle\phi|\psi\rangle|^2$. Although fidelity is a natural measure of state similarity, its quadratic form leads to three issues in NISQ inference:

1) **Sign Loss**: The absolute square operation $|\cdot|^2$ discards the sign of the real interference term.
2) **Non-linear Decision Boundaries**: Fidelity-induced decision boundaries are second-order surfaces in amplitude space.
3) **Shot Noise Sensitivity**: Estimating fidelity to high precision requires $O(1/\epsilon^2)$ measurement shots.

In contrast, the linear overlap $\langle\phi|\psi\rangle$ is complex-valued. We focus on its real part:

$$s(\psi) = \mathrm{Re}\langle\phi|\psi\rangle. \tag{2}$$

where $\mathrm{Re}\langle\phi|\psi\rangle$ denotes the real part of the inner product between the prototype state $|\phi\rangle$ and the input state $|\psi\rangle$.[1]

2) Decision Function and Labeling Rule: We define the binary classification decision based on the sign of the real overlap between the input state $|\psi\rangle$ and a predetermined class-representative state $|\phi\rangle$. The predicted label $\hat{y}$ is given by:

$$\hat{y} = \mathrm{sign}\left(\mathrm{Re}\langle\phi|\psi\rangle\right). \tag{3}$$

Setting the threshold at zero corresponds to a balanced classification. If prior class probabilities are known, a bias term $b \in \mathbb{R}$ can be added classically:

$$\hat{y} = \begin{cases} +1 & \mathrm{Re}\langle\phi|\psi\rangle + b \geq 0 \\ -1 & \mathrm{Re}\langle\phi|\psi\rangle + b < 0 \end{cases} \tag{4}$$

[1]Notation: $|\psi\rangle$ and $|\phi\rangle$ denote quantum states represented as complex vectors in Hilbert space. The expression $\langle\phi|\psi\rangle$ is their inner product, and $\mathrm{Re}\langle\phi|\psi\rangle$ is its real component. Symbols a_i and b_i denote amplitude coefficients of the corresponding state vectors.

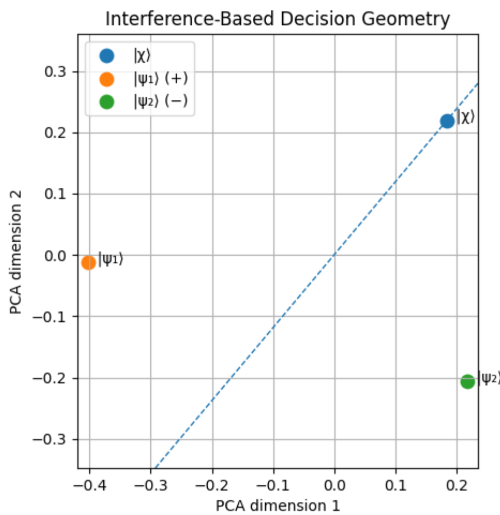

Fig. 3. Comparison of decision boundaries: Linear Interference (left) vs. Fidelity-based Quadratic (right). The interference rule $\mathrm{Re}\langle\phi|\psi\rangle = 0$ induces a linear hyperplane and retains the sign of the real interference term, while fidelity-based rules result in curved boundaries that discard sign information.

3) Decision Geometry: The Hilbert Space Hyperplane: Geometrically, the decision rule $\mathrm{Re}\langle\phi|\psi\rangle = 0$ defines a **linear hyperplane** that partitions the 2^n-dimensional Hilbert space into two half-spaces. Fig. 3 contrasts this linear-interference boundary with the curved, quadratic boundary induced by fidelity-based similarity, and Fig. 4 shows the fidelity geometry.

To see this, let $|\psi\rangle$ and $|\phi\rangle$ be represented by their amplitude vectors $\mathbf{c}_\psi, \mathbf{c}_\phi \in \mathbb{C}^{2^n}$. The real part of the inner product can be written as:

$$\mathrm{Re}\langle\phi|\psi\rangle = \frac{1}{2}\left(\langle\phi|\psi\rangle + \langle\psi|\phi\rangle\right) = \mathbf{Re}(\mathbf{c}_\phi^\dagger \mathbf{c}_\psi). \tag{5}$$

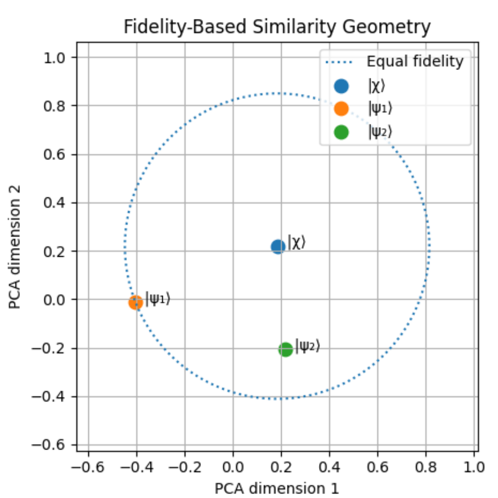

Fig. 4. Detailed geometric profile of fidelity-based similarity measures in Hilbert space, illustrating the nonlinear curvature of the decision surface.

4) Formal Derivation of the Decision Score: Consider the normalized state $|\psi\rangle = \sum_i a_i|i\rangle$ and the representative state $|\phi\rangle = \sum_i b_i|i\rangle$. The interference term $\mathrm{Re}\langle\phi|\psi\rangle$ can be expanded in terms of the real and imaginary parts of the coefficients:

$$\langle\phi|\psi\rangle = \sum_i b_i^* a_i = \sum_i (\mathrm{Re}\ b_i - i\mathrm{Im}\ b_i)(\mathrm{Re}\ a_i + i\mathrm{Im}\ a_i) \tag{6}$$

Taking the real part, we obtain:

$$\mathrm{Re}\langle\phi|\psi\rangle = \sum_i (\mathrm{Re}\ a_i \mathrm{Re}\ b_i + \mathrm{Im}\ a_i \mathrm{Im}\ b_i) \tag{7}$$

This summation is identical to the Euclidean inner product in $\mathbb{R}^{2\cdot 2^n}$. The decision boundary $\mathrm{Re}\langle\phi|\psi\rangle = 0$ is therefore the locus of all states in $\mathcal{H}$ that are "mutually orthogonal" under this real-part inner product.

E. Quantum Interference Circuit

The accessibility of the linear overlap signal $\mathrm{Re}\langle\phi|\psi\rangle$ is the defining hardware requirement of our framework. Since standard projective measurements $\langle\psi|O|\psi\rangle$ are quadratic in the state amplitudes, we utilize an ancilla-assisted interference circuit. In this framework, one inference corresponds to executing a fixed interference circuit on an input state and assigning a label based on the sign of the estimated overlap.

1) Physical Realization via the Hadamard Test: As with all gate-based quantum classifiers, state-preparation is treated as a separate cost; the inference complexity discussed here refers exclusively to the decision observable and measurement process.

The real part of the overlap between two quantum states can be directly extracted using the Hadamard test [18]. The circuit requires one additional ancilla qubit and the original register used to store the data states.

The procedure for evaluating $\mathrm{Re}\langle\phi|\psi\rangle$ is as follows:

1) **Initialization**: Prepare the ancilla qubit in $|0\rangle$ and the register in a state $|\psi\rangle$.
2) **First Hadamard**: Apply a Hadamard gate H to the ancilla qubit, putting it in the state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.
3) **Controlled Operation**: Apply a controlled unitary U that transforms $|\psi\rangle$ toward $|\phi\rangle$. In the most hardware-efficient setup, if we can prepare both $|\psi\rangle$ and $|\phi\rangle$ via unitaries V_ψ and V_ϕ from $|0\rangle$, the operation becomes a controlled version of $V_\phi V_\psi^\dagger$.
4) **Second Hadamard**: Apply a second Hadamard gate to the ancilla.
5) **Measurement**: Measure the ancilla qubit in the computational basis.

The probability of measuring the ancilla in state $|0\rangle$ is:

$$P(0) = \frac{1}{2} + \frac{1}{2}\mathrm{Re}\langle\phi|\psi\rangle. \tag{8}$$

The interference signal (our decision score) is then obtained from the expectation value of the ancilla's Z-observable:

$$\langle Z_{\text{anc}}\rangle = P(0) - P(1) = \mathrm{Re}\langle\phi|\psi\rangle. \tag{9}$$

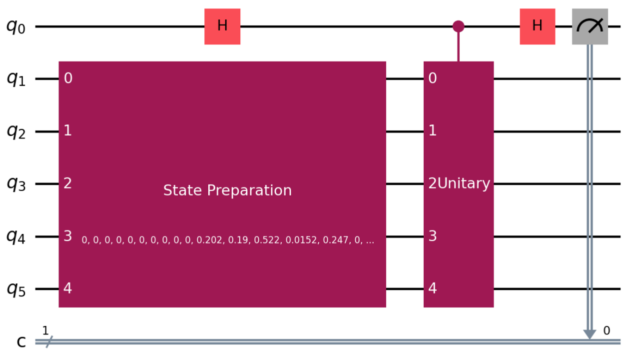

Fig. 5. Circuit diagram of the ISDO-B implementation. The controlled transition unitary $U_{\chi\psi}$ is used to estimate the linear overlap $\mathrm{Re}\langle\phi|\psi\rangle$ by measuring the ancilla qubit expectation value over a small constant number of shots.

2) ISDO-B: Hardware-Efficient Linear Interference: While the canonical Hadamard test relies on controlled state-preparation for both $|\psi\rangle$ and $|\phi\rangle$, the **ISDO-B (Interference-Based Static Decision Observable)** variant utilizes a controlled transition unitary $U_{\chi\psi}$ where $U_{\chi\psi}|\psi\rangle = |\phi\rangle$. We assume $U_{\chi\psi}$ is a state-transfer operator acting such that the target subspace is correctly mapped; its action on the orthogonal subspace is unconstrained as it does not contribute to the interference signal.

The circuit follows a modified Hadamard test structure:

1) Initialize ancilla in $|0\rangle$ and data register in $|0\rangle^{\otimes n}$.
2) Prepare test state $|\psi\rangle$ on the data register.
3) Apply Hadamard gate to the ancilla: $\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)\otimes|\psi\rangle$.
4) Apply controlled transition unitary $CU_{\chi\psi}$ targeting the data register: $\frac{1}{\sqrt{2}}(|0\rangle|\psi\rangle + |1\rangle|\phi\rangle)$.
5) Apply Hadamard gate to the ancilla and measure in the Z-basis.

The expectation value of the ancilla measurement yields $\langle Z_{\text{anc}}\rangle = \mathrm{Re}\langle\phi|\psi\rangle$. This formulation is particularly advantageous in static regimes where the class prototype $|\phi\rangle$ is precomputed, allowing for the engineering of specialized observables that avoid the overhead of full controlled state-preparation of arbitrary centroids.

3) Single-Circuit Inference Cost: The IQC uses a single fixed interference circuit per sample and reads out one ancilla qubit to obtain the decision sign. In our experiments, a small constant number of shots is sufficient to stabilize the sign decision, in contrast to fidelity-based methods that require high-precision probability estimation.

F. Class Representative State Construction

A pivotal design choice in the IQC framework is the complete decoupling of the learning process from quantum execution. Unlike VQCs, where learning involves an iterative quantum-classical loop, our framework performs all structural learning classically before anything is loaded onto the quantum processor.

1) Centroid-Based State Construction: The learning objective is to find a normalized quantum state $|\phi^{(c)}\rangle$ that best represents the distribution of training samples belonging to class $c \in \{-1, +1\}$. In this work, we adopt a **centroid-based aggregation rule**, which is conceptually equivalent to calculating the mean vector in Hilbert space. This centroid-based approach is functionally analogous to classical Linear Discriminant Analysis (LDA), where class means are used to establish a linear decision boundary, albeit implemented directly within the high-dimensional quantum Hilbert space.

Given a subset of training states $\mathcal{S}_c = \{|\psi_i\rangle : y_i = c\}$, where each $|\psi_i\rangle$ is an amplitude-encoded vector in $\mathbb{C}^d$, the unnormalized class-representative vector is:

$$|\tilde{\phi}^{(c)}\rangle = \sum_{|\psi_i\rangle \in \mathcal{S}_c} w_i |\psi_i\rangle, \tag{10}$$

where w_i are importance weights. In the simplest case, where all training samples are considered equally, $w_i = 1/|\mathcal{S}_c|$. The finalized class-representative state is obtained by normalization:

$$|\phi^{(c)}\rangle = \frac{|\tilde{\phi}^{(c)}\rangle}{\||\tilde{\phi}^{(c)}\rangle\|}. \tag{11}$$

Algorithm 1 details the classical preprocessing steps required to construct these states.

Algorithm 1 Static Learning of Class-Representative States

Require: Training dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$
1: Select feature map $\Phi : \mathbb{R}^d \to \mathbb{C}^{2^n}$ (e.g., Amplitude Encoding)
2: Initialize class accumulators: $\mathbf{v}_+ \leftarrow \mathbf{0}, \mathbf{v}_- \leftarrow \mathbf{0}$
3: **for** each $(\mathbf{x}_i, y_i) \in \mathcal{D}$ **do**
4: Compute state vector: $|\psi_i\rangle \leftarrow \Phi(\mathbf{x}_i)$
5: **if** $y_i = +1$ **then**
6: $\mathbf{v}_+ \leftarrow \mathbf{v}_+ + |\psi_i\rangle$
7: **else**
8: $\mathbf{v}_- \leftarrow \mathbf{v}_- + |\psi_i\rangle$
9: **end if**
10: **end for**
11: Normalize: $|\phi^{(+)}\rangle \leftarrow \mathbf{v}_+/\|\mathbf{v}_+\|, |\phi^{(-)}\rangle \leftarrow \mathbf{v}_-/\|\mathbf{v}_-\|$
12: **return** $|\phi^{(+)}\rangle, |\phi^{(-)}\rangle$

This aggregation is performed entirely in classical memory using the complex amplitude vectors of the data embeddings. For a dataset of size N and a feature space of dimension d, this classical step scales as $O(Nd)$, which is significantly faster than the $O(M \cdot N)$ scaling of VQC training (where M is the number of optimization iterations).

2) State Selection and Prototype Extraction: In some applications, a simple average may not be the most representative summary of a class, especially if the class distribution is multimodal. However, within the scope of this work, we assume a single-mode distribution for each class. The state $|\phi^{(c)}\rangle$ essentially acts as a "prototype" or "centroid" in the quantum feature space. The linear interference score $\mathrm{Re}\langle\phi|\psi\rangle$ can then be interpreted as the projection of the test sample $|\psi\rangle$ onto this class prototype.

3) Static Nature and Hybrid Efficiency: The resulting states $|\phi^{(+)}\rangle$ and $|\phi^{(-)}\rangle$ are static. Once computed, they are used to derive the control sequences for the interference circuit described earlier. There are no weight updates, no memory growth, and no backpropagation through quantum gates.

By treating the quantum device as a static inference engine rather than a trainable model, we eliminate several pathologies common in hybrid QML:

- **Gradient Noise**: Classical aggregation is deterministic and does not suffer from the sampling variance of quantum-derived gradients.
- **Barren Plateaus**: We avoid the optimization landscape altogether. The "training" is a simple linear aggregation with a guaranteed closed-form solution.
- **I/O Overhead**: The classical-quantum bridge is traversed only for data loading and the final inference shot, avoiding the high-latency loops of variational training.

V. IMPLEMENTATION

A. Dataset

The PatchCamelyon (PCam) dataset is a widely recognized benchmark for histopathological cancer detection, specifically designed for lymph node metastasis identification [13], [14], [16]. It consists of 96×96 RGB image patches extracted from whole-slide lymph node histology scans. Labels indicate whether tumor tissue appears in the central 32×32 region of the patch, yielding a binary task: metastatic (tumor) vs benign (non-tumor) [16]. These lymph node tissue samples are used for metastatic cancer detection. The task is binary classification of lymph node tissue patches into metastatic and benign categories. The full dataset contains 327,680 patches and is nearly class-balanced; the standard Kaggle split (duplicates removed) contains approximately 220k training patches and 57.5k test patches [14], [16].

Clinically, accurate detection of metastatic cancer in lymph nodes is crucial for cancer staging and treatment planning. The PCam dataset captures the morphological complexity of lymph node tissue, where malignant regions exhibit distinct cellular structures and tissue organization compared to benign areas. Fig. 6 shows representative examples of benign and malignant patches from the dataset, highlighting key histological differences relevant to classification.

B. CNN Feature Extraction

The CNN backbone and neck are trained classically using cross-entropy loss on PCam. We follow the original PCam train/val/test split provided by the dataset: the backbone is trained on the `train` split and validated on the `val` split for model selection. Due to simulation constraints and the compute-intensive nature of statevector inference, the `val` split (held out from training) is also used for the final downstream evaluation of the quantum stage, with the `test` split reserved for future hardware validation.

Training uses a lightweight PCamCNN with three convolutional blocks (3×3 conv + batch normalization + ReLU, with max-pooling after the first two blocks), followed by adaptive

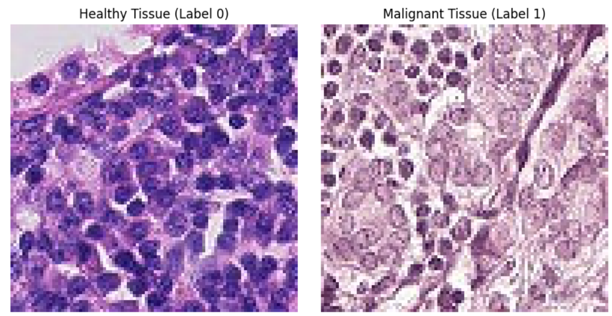

Fig. 6. Representative lymph node histopathology patches from the PCam dataset. Left: Benign tissue showing normal lymphatic architecture. Right: Malignant tissue exhibiting disrupted cellular morphology and tumor infiltration.

average pooling and a linear embedding head of dimension $d = 32$ with ReLU activation. A temporary linear classifier is attached during training and discarded after convergence. We use Adam (learning rate 10^{-3}, weight decay 10^{-4}), batch size 64, and a ReduceLROnPlateau scheduler (factor 0.5, patience 2) with early stopping (patience 10). Data augmentation is limited to label-preserving transforms: random horizontal/vertical flips and light color jitter, followed by normalization to mean $[0.5, 0.5, 0.5]$ and std $[0.5, 0.5, 0.5]$. Validation and embedding extraction use deterministic normalization only.

After training, the CNN+neck (PCamCNN) is frozen and used to produce embeddings for all training samples. These embeddings are then aggregated into class prototypes (Section IV-F). The quantum processor is used only at inference.

C. Quantum Encoding Setup

In our experiments, $d = 32$ and $n = 5$, so the quantum register consists of 5 data qubits and 1 ancilla qubit (6 total). Each embedding $\mathbf{x} \in \mathbb{R}^{32}$ is amplitude-encoded using Qiskit's `StatePreparation` unitary, which realizes the mapping $|\psi\rangle = \sum_{i=0}^{31} x_i |i\rangle$.

1) Depth Analysis of Amplitude Encoding: While the IQC inference circuit is fixed and short-depth (constant relative to state-preparation), the overall performance is bounded by the cost of preparing the states $|\psi\rangle$ and $|\phi\rangle$. For a d-dimensional input vector, amplitude encoding into $n = \log_2 d$ qubits requires a unitary U such that $U|0\rangle^{\otimes n} = \sum_{i=1}^{d} x_i |i\rangle$.

The standard decomposition of such a unitary using the Plesch-Brukner algorithm [12] results in a circuit with $O(d)$ CNOT gates and $O(d)$ single-qubit rotations. While this is exponential in the number of qubits n, it is linear in the number of features d. For NISQ deployment, one often uses approximate state-preparation techniques or hardware-efficient ansätze that trade off exact representation for reduced depth. In our simulation, we consider an exact decomposition but point out that the linear interference mechanism of the IQC is still valid independently of the state-preparation technique used, as long as the relative phase information is maintained.

D. Simulation Environment

Simulations are carried out via a statevector quantum simulator, with a tailored noise model added to simulate NISQ hardware characteristics. The shot complexity is measured relative to the physical measurement step, not simulation time. All results below were obtained in a single experimental run; multi-seed statistical validation is a planned future step.

All simulations were conducted using Python-based quantum and machine learning frameworks. Quantum circuits and statevector simulations were implemented using IBM's Qiskit framework, while classical model training and feature extraction were implemented using PyTorch. Experiments were executed on a workstation equipped with an AMD Ryzen 5 4600H CPU, an NVIDIA GTX 1650 GPU with 4 GB VRAM, and 24 GB of system memory. The GPU was primarily used for CNN training and embedding generation, while quantum circuit simulations were executed on the CPU.

- **Dataset**: PatchCamelyon (PCam), standard Kaggle split (duplicates removed) [16]. Due to statevector simulation memory constraints, we use a balanced held-out subset of **5000 samples** (2500 per class) drawn from the PCam validation partition.
- **Feature Extraction**: A trained PCamCNN backbone with a compact neck reduces each 96×96 RGB patch to an embedding of dimension $d = 32$. Embeddings are L2-normalized before quantum encoding.
- **Quantum Encoding**: Each embedding $\mathbf{x} \in \mathbb{R}^{32}$ is amplitude-encoded into $n = \lceil \log_2 32 \rceil = 5$ data qubits plus 1 ancilla (6 qubits total) using Qiskit's `StatePreparation` unitary, which realizes the mapping $|\psi\rangle = \sum_{i=0}^{31} x_i |i\rangle$.
- **Noise Model**: A depolarizing noise channel with error probability $p \in [0.01, 0.1]$ and thermal relaxation times (T_1, T_2) consistent with current superconducting processors.

E. Benchmark Models

Our IQC (ISDO-B variant) is compared to Fidelity-SWAP, Linear SVM, Logistic Regression, and k-NN ($k = 3$) on the same 5000-sample subset.

F. Evaluation Metrics

Fig. 7 compares classification accuracy between the proposed IQC, VQC, and fidelity-based methods across different noise levels.

Table II presents classification accuracy at a fixed noise level ($p = 0.01$).

TABLE II
CLASSIFICATION PERFORMANCE COMPARISON: VALIDATION OF IQC ACCURACY AGAINST QUANTUM AND CLASSICAL BASELINES UNDER SYMMETRIC DEPOLARIZING NOISE ($p = 0.01$).

Method	Accuracy (%)
IQC (Proposed)	**88.07**
Fidelity-SWAP	87.84
Linear SVM	90.53
Logistic Regression	90.47
k-NN ($k = 3$)	92.60

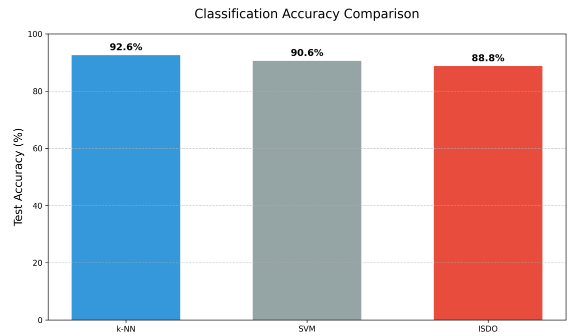

Fig. 7. Classification accuracy comparison between the proposed IQC, VQC, and Fidelity-based methods across different noise levels. The IQC maintains stable performance even as noise increases.

To validate shot-complexity behavior, we measured classification accuracy as a function of the number of measurement shots per sample.

TABLE III
INFERENCE ACCURACY VS. MEASUREMENT SHOTS: EMPIRICAL DEMONSTRATION OF MEASUREMENT-EFFICIENT BEHAVIOR FOR IQC RELATIVE TO FIDELITY AND VARIATIONAL METHODS.

Method	10 Shots	100 Shots	1000 Shots	4000 Shots
IQC (Proposed)	0.822	0.884	0.900	0.902
Fidelity-Based	0.654	0.814	0.906	0.906
VQC	0.808	0.808	0.808	0.808

VI. DISCUSSION

A. Measurement Efficiency

The IQC reaches near-peak performance with small constant shot counts, while fidelity-based and VQC models require substantially more measurements to stabilize classification scores. This reduces the duty cycle pressure on quantum hardware and improves throughput in measurement-limited settings.

The reduction in measurement shots has a direct impact on the energy consumption of the quantum-classical system. In superconducting architectures, each measurement shot involves microwave pulses and high-speed classical digitization, consuming significant power at the control electronics layer. By reducing the number of shots from 10^4 (typical for fidelity estimation) to a small constant number of shots (often $10^1 - 10^2$ for IQC sign estimation), we achieve a reduction in the "energy per inference" metric. This potentially positions interference-based QML as a more energy-efficient alternative for data processing in future quantum deployments.

B. Noise Robustness

Under realistic NISQ conditions, quantum circuits are subject to decoherence and operational errors. We analyze the sensitivity of the IQC decision signal to two primary noise channels.

1) Depolarizing Noise: A depolarizing channel $\mathcal{E}(\rho) = (1-p)\rho + p\frac{I}{2^{n+1}}$ maps a global state ρ of $n+1$ qubits to a mixture of itself and the maximally mixed state with probability p. For the ancilla-register system, this noise attenuates the interference signal by a factor $(1-p)^L$, where L is the circuit depth:

$$\langle Z_{\text{anc}} \rangle_{\text{noisy}} = (1-p)^L \text{Re}\langle \phi|\psi \rangle. \tag{12}$$

Crucially, since $(1-p)^L > 0$, the **sign** of the expectation value is preserved. The classification is correct as long as the shot noise does not exceed the attenuated signal.

2) Dephasing Noise: Dephasing or phase-damping errors $\mathcal{E}_{ph}(\rho) = (1-p)\rho + pZ\rho Z$ occur during the controlled unitary or idle times. In the Hadamard test, dephasing on the ancilla qubit adds a stochastic phase shift θ to the interference term:

$$P(0) = \frac{1}{2} + \frac{1}{2}\text{Re}(e^{i\theta}\langle \phi|\psi \rangle) \tag{13}$$

If the dephasing is symmetric around zero, then the expected decision score is unbiased. But if there is a systematic phase drift, then the decision boundary can be shifted. This implies that the ISDO-B is more robust to stochastic noise (such as depolarization) than to systematic calibration errors.

C. Limitations of the Model

A fundamental limitation of the current IQC framework is its reliance on a linear decision boundary in Hilbert space. While the Hilbert space dimension 2^n is large, separation depends heavily on the quality of the feature map $\Phi(\mathbf{x})$. If the data distribution is not linearly separable under the chosen map, the IQC behaves like a high-bias linear classifier. The dependence of inference stability on the classification margin mirrors that of classical linear classifiers and reflects intrinsic ambiguity near the decision boundary rather than a limitation of the interference-based inference mechanism.

All experiments reported in this work are conducted entirely via classical statevector simulation. No real quantum hardware experiments have been performed. Statevector simulation is noiseless by default; noise is injected via a synthetic depolarizing model. This means the reported results reflect idealized noise assumptions rather than the full complexity of real device errors (e.g., crosstalk, leakage, coherent errors). Furthermore, the current results come from a single experimental trial without multi-seed statistical validation. Real hardware validation on a backend such as IBM Quantum or IonQ, as well as multi-seed runs for mean $\pm$ std reporting, are identified as the most critical directions for strengthening the empirical claims of this work.

D. Scalability Considerations

The transition of the IQC from simulation to actual quantum hardware needs to take into account the issue of gate decomposition and ancilla usage. The Hadamard test needs controlled-unitary operations CU, which can be expensive in terms of circuit depth L when decomposed into native gates. The interference circuit relies on a single ancilla qubit that must be connected to all qubits in the data register used in the controlled operations; restricted connectivity can introduce SWAP overhead. Readout error rates can be as high as 1–5% in current NISQ devices, but the IQC's reliance on a single ancilla measurement helps isolate this effect. Readout error mitigation techniques can be applied to the ancilla qubit to improve the precision of the interference score without multi-qubit tomography.

The IQC framework exhibits favorable scaling compared to both classical linear models and variational quantum models. To represent a d-dimensional feature vector, the IQC requires $n = \lceil \log_2 d \rceil$ qubits, achieving logarithmic qubit scaling in representation (with linear gate complexity for state-preparation). Once the state is prepared, the interference circuit depth L is a constant or near-constant factor relative to the state-preparation depth. Inference uses a single fixed circuit and a small constant number of shots to stabilize the sign decision. The classical aggregation step scales as $O(Nd)$, which is optimal for linear learning. The reduction in measurement shots also lowers the energy per inference in measurement-limited hardware settings.

VII. Conclusion and Future Work

This paper proposes an interference-based quantum classifier that serves as a fixed inference primitive for medical image classification. By emphasizing the linear overlap $\text{Re}\langle \phi|\psi \rangle$ instead of quadratic probabilities, we achieve measurement-efficient inference and noise robustness. Our method separates the learning process from the quantum computation by calculating the class centroids classically. Experimental evaluation on PCam shows that our framework is comparable to variational models while being more efficient in terms of measurement cycles and optimization costs.

Future work will extend this framework to **adaptive learning**. We will study simple update rules for the class-representative states (e.g., exponential moving averages or small ensembles) and evaluate whether they improve separability without sacrificing the measurement efficiency of the linear interference core.

A. Extension to Multi-class Classification

While this work focuses on binary classification as the core primitive, the IQC framework can be extended to multi-class problems ($K > 2$) using standard decomposition strategies such as **One-vs-All (OvA)** or **One-vs-One (OvO)**.

In the OvA approach, for K classes, we learn K static class-representative states $\{|\phi^{(1)}\rangle, |\phi^{(2)}\rangle, \ldots, |\phi^{(K)}\rangle\}$. The interference operation is performed between the test state $|\psi\rangle$ and each class prototype. The classification rule is then:

$$\hat{y} = \arg\max_{c} \text{Re}\langle \phi^{(c)}|\psi \rangle. \tag{14}$$

The shot complexity for multi-class inference scales as $O(K)$, as we must perform K independent interference measurements. However, each measurement uses a small number of shots, so the overall efficiency remains significantly higher than multi-class VQCs or QSVMs. Future work will investigate more sophisticated ways to load multiple class prototypes into a single quantum memory for parallel interference, potentially reducing the scaling to $O(\log K)$.

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